Ozgur Ural, Ph.D.

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Machine Learning Research Scientist & Senior Software Engineer, Ph.D.

Machine learning research scientist and senior software engineer with more than twelve years of professional experience in industry and research, building production systems across cybersecurity, defense, and simulation, now leading enterprise AI strategy and applied GenAI at Avion. Completed a Ph.D. in Electrical Engineering and Computer Science at Embry-Riddle Aeronautical University in the United States (August 2025), with a machine-learning dissertation and three first-author IEEE Access publications: two on Proof-of-Learning and model watermarking, and a survey of blockchain-enhanced machine learning. My work centers on **trustworthy AI for mission-critical systems**: evidence of how a model was trained and who owns it, robustness against attackers, and predictable timing when a model runs inside a real-time loop.

TECHNICAL SKILLS

- **Languages:** C, C++, C#, Scala, Python, Go, Java, TypeScript, JavaScript
- **Systems and data:** multithreading, async I/O, low-latency, backpressure, Linux, networking, Boost, Qt/QML; gRPC, Protobuf, Redis, PostgreSQL, REST
- **Machine learning & AI:** PyTorch, TensorFlow, scikit-learn, deep learning, LLMs, retrieval-augmented generation (RAG), AI agents, on-prem and air-gapped model deployment, federated learning; trustworthy and verifiable ML (model watermarking, Proof-of-Learning, adversarial robustness, model provenance)
- **Web and DevOps:** Svelte, HTML, CSS, Tailwind; Docker, GitLab CI/CD, Jenkins, Git

EXPERIENCE

Avion Full Flight Simulators NIEUW-VENNEP, NETHERLANDS

Senior Software Engineer Oct 2023 – Present

- Lead Avion's enterprise AI strategy: authored the AI adoption roadmap and a three-tier architecture spanning air-gapped on-prem LLMs, a governed cloud-LLM API tier, and edge deployment.
- Won approval for two R&D projects under WBSO, the Dutch government's R&D tax-credit programme: sole person responsible for preparing and architecting both, for the Avion Cloud Environment and for autonomous engineering agents with verifiable oversight.
- Built Avion Assistant, an engineering-knowledge copilot offered to staff as a proof of concept: hybrid keyword and embedding retrieval with cross-encoder reranking, every answer cited to its source page, and golden-question checks on every release (Python, FastAPI, Qdrant, SvelteKit); added a prototype agent foundation that works with tools and records its actions.
- Designed the strategy's proposed AI workflows, each keeping a person accountable for the outcome, and built the strategy portal end to end (Flask, SQLite, Docker, GitLab CI/CD).
- Developed real-time simulator software in C, C++, and Scala, including a low-latency telemetry pipeline (JSON and Protobuf, Redis, PostgreSQL, gRPC) sustaining approximately 50 GB/s real-time throughput using batching and bounded-queue backpressure.
- Principal developer of the Avion Cloud Environment (about 70% of commits since 2024), a platform to configure, monitor and diagnose simulator bus systems (SvelteKit, TypeScript, Python, gRPC, Redis, PostgreSQL).
- Built the Avion Debriefing System: records a full-flight-simulator training session (displays, cockpit camera, instructor station, audio), then packages, encrypts and transfers it (C#/.NET 8, MSI installer); installed and verified on a simulator.
- Built NavDB Sync, version control for simulator navigation databases: operators keep their custom data through 28-day AIRAC updates, and merges run on the server so a conflict never reaches a simulator (Python, GitLab CI/CD).

Embry-Riddle Aeronautical University, Daytona Beach FLORIDA, USA

Graduate Research Assistant Sep 2021 – Dec 2024, May 2025 – Oct 2025

- Original research on proof-of-learning, model watermarking, and adversarial robustness, with three first-author papers in IEEE Access (2025 Journal Impact Factor 4.2, JCR 2026).

Havelsan ANKARA, TURKEY

Software Team Lead Nov 2020 – Aug 2021

Havelsan DLP enterprise Data Leakage Prevention product.

- Led a team of 14 engineers delivering endpoint security software; built its agent in C, C++, Java, and Go.

STM Defence Technologies ANKARA, TURKEY

Expert Software Engineer Feb 2019 – Nov 2020

Kargu Autonomous Rotary-Wing Attack UAV and Togan Autonomous Multi-Rotor Reconnaissance UAV.

- Developed critical components of the mission-control and ground-control systems in C++11 with Boost and Qt, under hard-real-time, safety-critical constraints.
- Designed and implemented Power-On and Continuous Built-In Tests for system reliability validation.

Comodo Cybersecurity ANKARA, TURKEY

Expert Software Engineer Jul 2014 – Feb 2019

Comodo Dome Secure Web Gateway, Comodo Patch Manager, and the Comodo Dragon browser.

- Led design and development of Windows services and applications in C++17 with Boost and WinAPI; established unit testing and code review, and CI/CD pipelines with Jenkins.

EDUCATION

Embry-Riddle Aeronautical University, Daytona Beach FLORIDA, USA

Ph.D. in Electrical Engineering and Computer Science Aug 2021 – Aug 2025

- Dissertation: Enhancing Proof-of-Learning Security Against Spoofing Attacks Using Model Watermarking. Advisor: Dr. Kenji Yoshigoe; committee included IEEE Fellow Dr. Houbing Song.

Middle East Technical University (METU) ANKARA, TURKEY

Master of Science in Cyber Security 2015 – 2019

- Thesis: Automatic Detection of Cyber Security Events from Turkish Twitter Stream and Newspaper Data. Advisor: Prof. Cengiz Acartürk.

Middle East Technical University (METU) ANKARA, TURKEY

Bachelor of Science in Computer Engineering 2010 – 2014

- Turkey's most selective computer engineering program (English-medium), entered among roughly the top 1,000 of more than 2 million national candidates. Graduated with First Prize in TUBITAK's national project competition and Second Prize at the METU Computer Engineering graduation Demo Day (2014).

Anadolu University ESKISEHIR, TURKEY

Bachelor of Business Administration 2012 – 2017

PUBLICATIONS

- Ural, O. and Yoshigoe, K. (2025). SecurePoL: Integration of Watermarking With Proof-of-Learning to Enhance Security Against Spoofing Attacks. *IEEE Access*, vol. 13, pp. 213067–213091.
- Ural, O. (2025). Enhancing Proof-of-Learning Security Against Spoofing Attacks Using Model Watermarking. Doctoral dissertation, Embry-Riddle Aeronautical University.
- Ural, O. and Yoshigoe, K. (2024). Enhancing Security of Proof-of-Learning Against Spoofing Attacks Using Feature-Based Model Watermarking. *IEEE Access*, vol. 12, pp. 169567–169591.
- Ural, O. and Yoshigoe, K. (2023). Survey on Blockchain-Enhanced Machine Learning. *IEEE Access*, vol. 11, pp. 145331–145362.
- Ural, O. and Acartürk, C. (2021). Automatic Detection of Cyber Security Events from Turkish Twitter Stream and Newspaper Data. Proceedings of the 7th International Conference on Information Systems Security and Privacy (ICISSP), pp. 66–76. DOI: 10.5220/0010201600660076.
- Ural, O. and Erdur, E. (2016). Secure Proxy on Cloud. Technical report. DOI: 10.13140/RG.2.2.24058.08649.

For full list and citation metrics: Google Scholar profile. Narrated explainers of this research: ozgurural.github.io/lab.

PROFESSIONAL SERVICE

- **Reviewer:** IEEE Access and other journals, IEEE and ACM conferences (security, ML, distributed systems).
- **Program Committee (2026):** NLPAICS 2026, Workshop on NLP Applied to Information and Cyber Security, Alicante.